# Riemann Hypothesis via Warped Stationarity — Proof DAG

Stephen Crowley

2026-04-22

**Abstract**

Every nontrivial zero of $\zeta$ lies on $\Re s = \frac{1}{2}$.

## Axioms

**Axiom 1** (Hardy Z-function)**.** $Z(t) \coloneqq e^{i\vartheta(t)}\,\zeta(\frac{1}{2}+it)$, where $\vartheta(t) \coloneqq \arg\Gamma(\frac{1}{4} + it/2) - (t/2)\log\pi$. $Z$ is real on $\mathbb{R}$; $|Z(t)| = |\zeta(\frac{1}{2}+it)|$; zeros of $Z$ on $\mathbb{R}$ coincide with zeros of $\zeta$ on $\frac{1}{2}+i\mathbb{R}$.

*Source.* Edwards, Riemann's Zeta Function, §6.5; Titchmarsh Ch. 2

**Axiom 2** (Functional equation symmetry)**.** $\xi(w) \coloneqq \frac{1}{2}\,w(w-1)\,\pi^{-w/2}\,\Gamma(w/2)\,\zeta(w)$ is entire and satisfies $\xi(w) = \xi(1-w)$. Consequently, if $\rho = \beta+i\gamma$ is a nontrivial zero with $\beta \in (0,1)$, $\gamma \in \mathbb{R}$, then $1-\bar{\rho} = (1-\beta)+i\gamma$ is also a nontrivial zero.

*Source.* Titchmarsh, The Theory of the Riemann Zeta-Function, §2.12

**Axiom 3** (Holomorphy of $\vartheta$ on $|\Im t| < \frac{1}{2}$)**.** $\vartheta(t) = \frac{1}{2}\operatorname{Im}\log\Gamma(\frac{1}{4}+it/2) - (t/2)\log\pi$ extends to a holomorphic function on $\Omega \coloneqq \{t \in \mathbb{C} : |\Im t| < \frac{1}{2}\}$. The poles of $\psi(\frac{1}{4}+it/2) = \Gamma'/\Gamma(\frac{1}{4}+it/2)$ occur where $\frac{1}{4}+it/2 \in \{0, -1, -2, \ldots\}$; equivalently $t \in \{i(\frac{1}{2}+2k) : k \geq 0\} \cup \{-i(\frac{1}{2}+2k) : k \geq 0\}$, all with $|\Im t| \geq \frac{1}{2}$.

*Source.* Abramowitz–Stegun 6.3.5; direct computation from $\Gamma$

**Axiom 4** (Unique pole of $\zeta$)**.** $\zeta$ is meromorphic on $\mathbb{C}$ with a simple pole at $w = 1$ and holomorphic elsewhere. For $t \in \Omega$, $\frac{1}{2}+it = 1$ iff $t = -i/2$, which lies on $\partial\Omega$, not in $\Omega$; hence the map $t \mapsto \zeta(\frac{1}{2}+it)$ is holomorphic on $\Omega$.

*Source.* Titchmarsh, §2.1

**Axiom 5** (Digamma asymptotic)**.** As $|t| \to \infty$ with $t \in \mathbb{R}$, $\frac{1}{2}\Re\psi(\frac{1}{4}+it/2) = \frac{1}{2}\log|t/2| + \mathcal{O}(|t|^{-2})$. Consequently $\vartheta'(t) = \frac{1}{2}\log|t/(2\pi)| + \mathcal{O}(|t|^{-2}) \to +\infty$.

*Source.* Olver, Asymptotics and Special Functions, Ch. 8 §4

**Axiom 6** (Hardy–Littlewood second moment)**.** $\int_0^T |\zeta(\frac{1}{2}+it)|^2\,dt = T\log(T/(2\pi)) + (2\gamma_e - 1)\,T + \mathcal{O}(T^{1/2}\log T)$, where $\gamma_e$ is the Euler–Mascheroni constant.

*Source.* Titchmarsh, Theorem 7.3

**Axiom 7** (Riemann–Siegel formula)**.** For $t > 0$: $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$, where $N(t) \coloneqq \lfloor\sqrt{t/(2\pi)}\rfloor$ and $R(t) = \mathcal{O}(t^{-1/4})$ with $(d/dt) \arg R(t) = \mathcal{O}(t^{-1/2})$.

*Source.* Edwards §7.4; Gabcke (1979) Satz 7.1; Berry (1995) Proc. Roy. Soc. A 450

**Axiom 8** (Wiener's Theorem 29)**.** For $f \in L^p_{loc}(\mathbb{R})$ with $p = 2$ and time-average covariance $c_F(\eta) \coloneqq \lim_{U\to\infty} U^{-1} \int_0^U f(u)\, \bar{f}(u+\eta)\, du$ existing, positive-definite, and continuous at 0, the Bochner spectral measure $\mu[f]$ defined by $c_F(\eta) = \int e^{i\eta\beta}\, d\mu[f](\beta)$ satisfies $\mu[f](E) = \lim_{T\to\infty} (2\pi/T) \int_e |(2\pi)^{-1} \int_0^T f(u)\, e^{-imu}\, du|^2\, dm$ for every bounded Borel set $E \subset \mathbb{R}$.

*Source.* Wiener, Acta Math. 55 (1930), Theorem 29

**Axiom 9** (Paley–Wiener for WSS functions with compactly supported spectrum)**.** If $f \in L^2_{loc}(\mathbb{R})$ has Bochner spectral measure $\mu[f]$ with $\operatorname{supp} \mu[f] \subseteq [-\tau, \tau]$, then $f$ extends to an entire function on $\mathbb{C}$ of exponential type $\le \tau$ satisfying $|f(z)| \le \sqrt{(\mu[f](\mathbb{R}))} \cdot e^{t|\Im z|}$.

*Source.* Yaglom, Correlation Theory, Vol. I, Thm. 4.7.3; Bochner–Phillips, Ann. Math. 43 (1942), 409–418

**Axiom 10** (Akhiezer–Laguerre–Pólya criterion)**.** If $F : \mathbb{C} \to \mathbb{C}$ is an entire function of exponential type $\le \tau$ which is real on $\mathbb{R}$ and whose autocorrelation measure $\mu[F]$ (from $c[F]$) is finite, non-negative, and supported in $[-2\tau, 2\tau]$, then $F \in \mathscr{LP}$ (Laguerre–Pólya class) and every zero of $F$ is real.

*Source.* Akhiezer, Theory of Approximation, §V.4; Levin, Lectures on Entire Functions, Lecture 16, Theorem 3

## Definitions

**Definition 11** (Shifted phase $\Theta$)**.** $\Theta(t) \coloneqq \vartheta(t) + c\, t$, where $c$ is a real parameter with $c > c_\star \coloneqq -\inf \vartheta'(\mathbb{R})$ (i.e. $c$ exceeds minus the infimum of $\vartheta'$ over $\mathbb{R}$).

**Definition 12** (Unwarped function Y)**.** $Y : \mathbb{R} \to \mathbb{R}$, $Y(u) \coloneqq Z(\Theta^{-1}(u)) / \sqrt{(\Theta'(\Theta^{-1}(u)))}$.

**Definition 13** (Windowed Fourier K[T])**.** (μ)] $K[T](\mu) \coloneqq (2\pi)^{-1} \int_0^T Z(t) \sqrt{(\Theta'(t))}\, e^{-im\theta(t)}\, dt = (2\pi)^{-1}$ ∫ from 0 to $\Theta(T)$ $Y(u)\, e^{-imu}\, du$.

## Lemmas

**Lemma 14** ($c_\star$ is finite)**.** *$c_\star \coloneqq -\inf \vartheta'(\mathbb{R})$ lies in $(0, \infty)$.*

*Proof.* $\vartheta'$ is continuous on $\mathbb{R}$. By ax:ψasymp, $\vartheta'(t) \to +\infty$ as $|t| \to \infty$. A continuous function on $\mathbb{R}$ tending to $+\infty$ at infinity attains a finite infimum. □

**Lemma 15** ($\Theta$ is a $C^\infty$ bijection $\mathbb{R} \to \mathbb{R}$ with $\Theta' > 0$)**.** *For $c > c_\star$, $\Theta$ is smooth with $\Theta'(t) = \vartheta'(t) + c > 0$ for every $t \in \mathbb{R}$; $\Theta : \mathbb{R} \to \mathbb{R}$ is a $C^\infty$ diffeomorphism.*

*Proof.* $c > c_\star = -\inf \vartheta'$ gives $\vartheta' + c > 0$ on $\mathbb{R}$. Strict monotonicity combined with $\Theta(\pm\infty) = \pm\infty$ gives bijectivity. The inverse function theorem gives smoothness of $\Theta^{-1}$. □

**Lemma 16** (Pointwise factorization $Z = \sqrt{\Theta'} \cdot Y \circ \Theta$). *For every $t \in \mathbb{R}$, $Z(t) = \sqrt{(\Theta'(t))} \cdot Y(\Theta(t))$.*

*Proof.* Set $u \coloneqq \Theta(t)$. Then $\Theta^{-1}(u) = t$ and $\Theta'(\Theta^{-1}(u)) = \Theta'(t)$. By def:Y, $Y(\Theta(t)) = Y(u) = Z(t) / \sqrt{(\Theta'(t))}$. Multiply both sides by $\sqrt{(\Theta'(t))}$. □

**Lemma 17** ($\vartheta'' > 0$ on $(0, \infty)$). *$\vartheta''(t) > 0$ for every $t > 0$. In particular, $\vartheta'$ is strictly increasing on $(0, \infty)$.*

*Proof.* $\vartheta''(t) = -\frac{1}{4} \Im\, \psi'(\frac{1}{4}+it/2)$. The series $\psi'(w) = \sum_{k\geq 0} (w+k)^{-2}$ gives, for $w = \frac{1}{4}+it/2$ with $t > 0$, $\Im(w+k)^{-2} = -2(\frac{1}{4}+k)(t/2)\, |w+k|^{-4} < 0$. Hence $-\frac{1}{4} \Im\psi'(\frac{1}{4}+it/2) > 0$. □

**Lemma 18** (Instantaneous frequency ratio $\omega_n \in [0, 1)$). *For every $n \geq 1$, every $c > c\star$, and every $t \geq 2\pi\, n^2$, define $\omega_n(t) \coloneqq (\vartheta'(t) - \log n) / (\vartheta'(t) + c)$. Then $\omega_n(t) \in [0, 1)$.*

*Proof.* By Edwards §7.4 (stationary-phase saddle at $t = 2\pi\, n^2$), $\vartheta'(2\pi\, n^2) = \log n$. By lem:$\vartheta''$, $\vartheta'$ is strictly increasing on $(0, \infty)$, so $\vartheta'(t) \geq \log n$ for every $t \geq 2\pi\, n^2$; hence the numerator is $\geq 0$. By lem:Θdiffeo, the denominator $\vartheta'(t) + c$ is $> 0$. Denominator minus numerator equals $c + \log n > 0$, so $\omega_n(t) < 1$. □

## Theorems and Corollaries

**Theorem 19** (Time-average WSS of Y). *The time-average covariance $c[Y](\eta) \coloneqq \lim_{U\to\infty} U^{-1} \int_0^U Y(u)\, Y(u+\eta)\, du$ exists for every $\eta \in \mathbb{R}$, is positive-definite, and is continuous at 0. In particular $c[Y](0) = 2$.*

*Proof.* Change of variable $u = \Theta(t)$, $du = \Theta'(t)\, dt$ gives $\int_0^U Y(u)^2\, du = \int$ from 0 to $\Theta^{-1}(U)$ $|Z(t)|^2\, dt$. By ax:HL, $\int_0^T |Z(t)|^2\, dt = T \log(T/(2\pi)) + \mathcal{O}(T)$. By lem:Θdiffeo and ax:ψasymp, $U = \Theta(T) = \vartheta(T) + c\, T = (T/2) \log(T/(2\pi)) + \mathcal{O}(T)$, hence $U^{-1} \int_0^U Y^2\, du \to 2$. For general $\eta$, apply ax:RS to write $Y(\Theta(t))\, Y(\Theta(t)+\eta) = (\Theta'(t)\, \Theta'(t + \delta[\eta](t)))^{-1/2}\, Z(t)\, Z(t + \delta[\eta](t))$, where $\delta[\eta](t) \coloneqq \Theta^{-1}(\Theta(t)+\eta) - t = \eta / \Theta'(\tau)$ for some $\tau$ between $t$ and $t + \delta[\eta](t)$ (mean value theorem). Split $Z(t)\, Z(t + \delta[\eta](t))$ into diagonal ($m = n$, $\sigma = \sigma'$), antidiagonal ($m = n$, $\sigma = -\sigma'$), off-diagonal ($m \neq n$), and remainder cross-terms. The diagonal term $(n, \sigma)$ contributes $(1/U) \sum_{n\leq n(t)} (1/n) \int \cos(\eta\, \omega_n(t))\, dt$, which converges by lem:ωbound and bounded convergence to a finite limit $A(\eta)$. The antidiagonal, off-diagonal, and remainder terms are $\mathcal{O}(U)$ times $o(1)$ by integration-by-parts applied to their oscillatory phase slopes ($\geq 2\vartheta'(t) - \mathcal{O}(1)$ for $\sigma \neq \sigma'$, and $\geq \frac{1}{2}\, |\log(m/n)|$ for $m \neq n$), together with Cauchy–Schwarz on the R cross-terms. Positive-definiteness: $c[Y]$ is a Cesàro limit of positive-definite kernels, which is positive-definite. Continuity at 0: $|c[Y](\eta) - c[Y](0)|^2 \leq c[Y](0) \cdot (c[Y](0) - \Re\, c[Y](\eta)) \to 0$ via the explicit diagonal series. □

**Corollary 20** (Bochner spectral measure $\mu[Y]$). *] There is a finite non-negative Borel measure $\mu[Y]$ on $\mathbb{R}$ with $c[Y](\eta) = \int e^{i\eta\beta}\, d\mu[Y](\beta)$ and $\mu[Y](\mathbb{R}) = c[Y](0) = 2$.*

*Proof.* Bochner's theorem applied to $c[Y]$ (positive-definite and continuous at 0 by thm:WSS). □

**Theorem 21** (Band-limit). *For every $c > c\star$ and every $\mu \in \mathbb{R}$ with $|\mu| > 1$, $\lim_{T\to\infty} 2\pi\, |K[T](\mu)|^2 / \Theta(T) = 0$.*

*Proof.* Insert ax:RS into def:K[T] and substitute $u = \Theta(t)$. The $(n, \sigma)$-mode of the main sum contributes $\mathscr{J}_{n,\sigma}(T, \mu) \coloneqq \int$ from $u_n$ to $\Theta(T)$ $G_{n,\sigma}(u)\, e^{iP(u)}\, du$, where $u_n \coloneqq \Theta(2\pi n^2)$, $G_{n,\sigma}(u) = x^{-1/2}$ with $x \coloneqq \Theta'(\Theta^{-1}(u))$, and $P'(u) \coloneqq \sigma\, \omega_n(\Theta^{-1}(u)) - \mu$. By lem:ωbound, $|P'(u)| \geq |\mu| - 1 =: \lambda > 0$ for $|\mu| > 1$. Write $Q_{n,\sigma,\mu}(u) \coloneqq G_{n,\sigma}(u) / P'(u) = x^{1/2} / \hat{P}(x)$, with $\hat{P}(x) \coloneqq (\sigma - \mu)\, x - \sigma\, (c + \log n)$; Q is a rational function of x alone (no $\vartheta''$ enters). A single integration by parts gives $\mathscr{J}_{n,\sigma} = Q \cdot e^{iP} / i$ evaluated from $u_n$ to $\Theta(T) - \int Q'(u)\, e^{iP(u)} / i\, du$, so $|\mathscr{J}_{n,\sigma}| \leq 2 \sup |Q| + \int |Q'(u)|\, du$. On $[u_n, \infty)$, $x \geq \beta_n \coloneqq \Theta'(2\pi n^2)$ and $|\hat{P}(x)| \geq \lambda\, x$, hence $|Q(u)| \leq \lambda^{-1}\, \beta_n^{-1/2}$. For the derivative, $dQ/dx = (-(\sigma - \mu)\, x - 2\sigma\, (c + \log n)) / (2\, x^{1/2}\, \hat{P}(x)^2)$, and the bound $|dQ/dx| \leq ((|\mu|+1) / (2\lambda^2))\, x^{-3/2} + ((c + \log n) / \lambda^2)\, x^{-5/2}$ follows from $|\hat{P}(x)| \geq \lambda\, x$. Change of variable $u \mapsto x$ with $du/dx = 1/\vartheta''(\Theta^{-1}(u)) > 0$ (lem:ϑ″) yields $\int$ from $u_n$ to $\infty$ $|Q'(u)|\, du = \int$ from $\beta_n$ to $\infty$ $|dQ/dx|\, dx \leq ((|\mu|+1) / \lambda^2)\, \beta_n^{-1/2} + (2(c + \log n) / (3\lambda^2))\, \beta_n^{-3/2}$. Since $\beta_n \geq c + \log n$, the $\beta_n^{-3/2}$ term is $\leq (2 / (3\lambda^2))\, \beta_n^{-1/2}$. Therefore $|\mathscr{J}_{n,\sigma}(T, \mu)| \leq C_1(c, |\mu|)\, \beta_n^{-1/2}$ with $C_1(c, |\mu|) \coloneqq 2\lambda^{-1} + (|\mu|+1)\, \lambda^{-2} + (2/3)\, \lambda^{-2}$. The Riemann–Siegel remainder contributes $\int R(t)\, \sqrt{\Theta'(t)}\, e^{-im\theta(t)}\, dt$; by ax:RS, $(d/dt) \arg R(t) = \mathcal{O}(t^{-1/2})$, so for large t the total phase slope $\mu\, \Theta'(t) + (d/dt) \arg R(t)$ is $\geq |\mu|\, c / 2$, and integration by parts yields a uniform $\mathcal{O}(T^{-1/4})$ contribution. Summing over $n \leq N(T) = \mathcal{O}(T^{1/2})$: $|K[T](\mu)| \leq C_1 \sum_{n=1}^{N(T)} n^{-1/2}\, \beta_n^{-1/2} + \mathcal{O}(T^{-1/4}) = \mathcal{O}(T^{1/4} / \sqrt{\log T})$ (using $\beta_n \geq c + \log n$). Since $\Theta(T) \sim (T/2) \log T$, $2\pi\, |K[T](\mu)|^2 / \Theta(T) = \mathcal{O}(T^{-1/2} / (\log T)^2) \to 0$. □

**Corollary 22** (supp μ[Y]. ⊆ *[−1, 1]] The Bochner spectral measure μ[Y] is supported in [−1, 1].*

*Proof.* By ax:Wiener29, $\mu[Y](E) = \lim_{T\to\infty} (2\pi / \Theta(T)) \int_E |K[T](m)|^2\, dm$ for every bounded Borel $E \subset \mathbb{R}$. For $E \subset \{\!|\, |m| > 1 \,|\!\}$, thm:bandlim gives pointwise vanishing of the integrand, and dominated convergence (the integrand is uniformly bounded) gives $\mu[Y](E) = 0$. Countable additivity extends to $\mu[Y](\{\!|\, |m| > 1 \,|\!\}) = 0$. □

**Theorem 23** (Y extends to an entire function of exponential type ≤ 1)**.** *Y extends to an entire function on $\mathbb{C}$ with $|Y(z)| \leq \sqrt{2} \cdot e^{|\Im z|}$ for every $z \in \mathbb{C}$.*

*Proof.* Apply ax:PW with $\tau = 1$ and $\mu[f] = \mu[Y]$. The hypotheses are: supp $\mu[Y] \subseteq [-1, 1]$ (cor:supp) and $\mu[Y](\mathbb{R}) = 2$ (cor:Bochner). The conclusion gives $|Y(z)| \leq \sqrt{2} \cdot e^{|\Im z|}$. □

**Theorem 24** (All zeros of Y are real)**.** *Every zero $z \in \mathbb{C}$ of the entire extension of Y satisfies $z \in \mathbb{R}$.*

*Proof.* Y is real on $\mathbb{R}$ (def:Y). By thm:Yentire, Y is entire of exponential type ≤ 1. By cor:supp, μ[Y] is supported in $[-1, 1] \subseteq [-2, 2]$. Apply ax:Akhiezer with $\tau = 1$: $Y \in \mathscr{L}\mathscr{P}$ and every zero of Y is real. □

**Theorem 25** (Z is holomorphic on $\Omega = \{\!|\, |\Im t| < ½ \,|\!\}$)**.** *$Z(t) = e^{i\vartheta(t)}\, \zeta(½+it)$ extends to a holomorphic function on $\Omega \coloneqq \{\!|\, t \in \mathbb{C} : |\Im t| < ½ \,|\!\}$.*

*Proof.* By ax:ϑhol, ϑ is holomorphic on Ω. By ax:ζpole, ζ has a unique pole at $w = 1$, which corresponds to $t = -i/2 \in \partial\Omega \setminus \Omega$; hence $\zeta(½+it)$ is holomorphic on Ω. Products and compositions of holomorphic functions are holomorphic. □

**Theorem 26** (Riemann Hypothesis)**.** *Every nontrivial zero of ζ satisfies $\Re\rho = ½$.*

*Proof.* Let $\rho = \beta + i\gamma$ be a nontrivial zero with $\beta \in (0, 1)$ and $\gamma \in \mathbb{R}$. By ax:ζFE, $\rho \mapsto 1 - \bar{\rho}$ is a symmetry, so it suffices to treat $\beta \in (0, ½]$. Assume for contradiction $\beta < ½$, and set $d \coloneqq ½ - \beta \in (0, ½)$.

(i) The segment $S \coloneqq \{\!| \gamma + is : s \in [0, d] |\!\}$ is compact and $S \subset \Omega$. By ax:ϑhol, $\vartheta'$ is continuous on $S$; define $M \coloneqq -\inf \{\!| \mathfrak{R}\, \vartheta'(\gamma + is) : s \in [0, d] |\!\} \in \mathbb{R}$. Fix $c > \max(c\star, M)$; then $c - M > 0$ and $\mathfrak{R}\, \Theta'(\gamma + is) = \mathfrak{R}\, \vartheta'(\gamma + is) + c \geq c - M > 0$ for every $s \in [0, d]$.

(ii) Let $N \coloneqq \{\!| s \in \mathbb{C} : |\mathfrak{R}s| < ½ |\!\}$, an open connected subset of $\mathbb{C}$. For every $s \in N$, $\gamma + is \in \Omega$, so $Z(\gamma + is)$, $\vartheta(\gamma + is)$, $\Theta(\gamma + is) = \vartheta(\gamma + is) + c\,(\gamma + is)$, and $\Theta'(\gamma + is) = \vartheta'(\gamma + is) + c$ are holomorphic in $s$ (thm:Zhol, ax:ϑhol). By thm:Yentire, $Y$ is entire. Define $F, G : N \to \mathbb{C}$ by $F(s) \coloneqq Z(\gamma + is)$ and $G(s) \coloneqq \sqrt{(\Theta'(\gamma + is))} \cdot Y(\Theta(\gamma + is))$, with $\sqrt{}$ the principal branch (cut along $(-\infty, 0]$).

(iii) Let $N'$ be the connected component of $\{\!| s \in N : \Theta'(\gamma + is) \notin (-\infty, 0] |\!\}$ containing $[0, d]$. By (i), $\mathfrak{R}\, \Theta' > 0$ on $[0, d]$, so $\Theta'(\gamma + is)$ lies in the open right half-plane there, which is a subset of $\mathbb{C} \setminus (-\infty, 0]$; continuity of $\Theta'$ extends this to an open neighborhood of $[0, d]$ in $\mathbb{C}$, hence $[0, d] \subset N'$. Since $0 \in N'$ and $N'$ is open, there exists $\varepsilon > 0$ such that $\{\!| ir : |r| < \varepsilon |\!\} \subset N'$. $F$ and $G$ are holomorphic on $N'$.

(iv) For $s = ir$ with $|r| < \varepsilon$, $\gamma + is = \gamma + i(ir) = \gamma - r \in \mathbb{R}$. By lem:Zfact applied at $\gamma - r$, $F(ir) = Z(\gamma - r) = \sqrt{(\Theta'(\gamma - r))} \cdot Y(\Theta(\gamma - r)) = G(ir)$.

(v) $F - G$ is holomorphic on $N'$ and vanishes on $\{\!| ir : |r| < \varepsilon |\!\}$, which has accumulation points in $N'$. The identity theorem gives $F \equiv G$ on $N'$. In particular, $s = d \in N'$, and $F(d) = G(d)$: $Z(\gamma + id) = \sqrt{(\Theta'(\gamma + id))} \cdot Y(\Theta(\gamma + id))$.

(vi) $½ + i(\gamma + id) = ½ + i\gamma - d = \beta + i\gamma = \rho$, and $\gamma + id \in \Omega$ (since $\mathfrak{I}(\gamma + id) = d \in (0, ½)$), hence $Z(\gamma + id) = e^{iv(\gamma+id)} \cdot \zeta(\rho) = 0$ (thm:Zhol, ax:Z, $\rho$ a zero of $\zeta$).

(vii) $\Theta'(\gamma + id) \notin (-\infty, 0]$ (since $d \in N'$), so $\sqrt{(\Theta'(\gamma + id))} \neq 0$. From (v)–(vi), $Y(\Theta(\gamma + id)) = 0$. By thm:Yzeros, $\Theta(\gamma + id) \in \mathbb{R}$.

(viii) $\Theta(\gamma + is)$ is holomorphic in $s$ on $N'$; parametrize the segment $s \in [0, d]$: $\Theta(\gamma + id) - \Theta(\gamma) = \int_0^d (d/ds)\, \Theta(\gamma + is)\, ds = \int_0^d i \cdot \Theta'(\gamma + is)\, ds$. Take imaginary parts; $\Theta(\gamma) \in \mathbb{R}$ (since $\gamma \in \mathbb{R}$): $\mathfrak{I}\, \Theta(\gamma + id) = \int_0^d \mathfrak{R}\, \Theta'(\gamma + is)\, ds \geq d \cdot (c - M)$.

(ix) By (vii), $\mathfrak{I}\, \Theta(\gamma + id) = 0$. By (i), $c - M > 0$. By (viii), $0 \geq d\,(c - M)$. Therefore $d = 0$, i.e. $\beta = ½$, contradicting $\beta < ½$. Hence $\beta = ½$. □